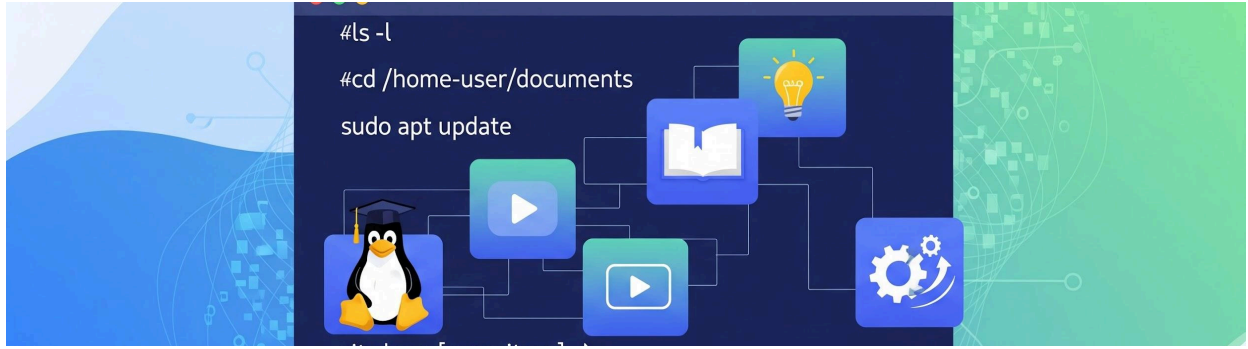




# LINUX OS

## **Introduction:**

- Linux is a free and open-source operating system with high security.
- Linux is a multi-user operating system.

## **OPERATING SYSTEM**

- An Operating System (OS) is software that acts as an interface between computer hardware components and the user.
- Every computer system must have at least one operating system to run other programs.
- Applications like browsers, MS Office, Notepad, games, etc., need an environment to run and perform tasks.
- The OS helps users communicate with the computer without knowing the computer's language.

## **KERNEL**

- The kernel manages the hardware components such as CPU, memory, and peripheral devices.
- The kernel is the lowest level of the operating system.

## **DAEMONS**

- Daemons manage background services such as printing, sound, scheduling, etc.
- These services start during system boot or after user login.

## **SHELL**

- The shell is an environment where users can run commands, programs, and shell scripts.
- It takes input from the user, executes it, and displays the output.
- A shell acts as an interface between the user and the operating system.

Shells can be:

Ø Graphical (GUI-based)

Ø Command-line based (CLI)

## **COMMAND**

- A command is an instruction or request given to the operating system by a user.
- It tells the computer to perform a specific task.

## **TERMINAL**

- A terminal is a text-based interface that allows users to interact with the operating system by typing commands and receiving output.
- Terminals are commonly used in Linux, Unix, macOS, and Windows systems.

## **LINUX OS DISTRIBUTIONS**

- Linux distributions are modified versions of Linux released with different names based on user requirements.

Examples:

- i. RedHat
- ii. Ubuntu
- iii. Debian
- iv. CentOS
- v. Fedora
- vi. OpenSUSE
- vii. Kali Linux
- viii. Amazon Linux

## ix. Rocky Linux

### **LINUX HISTORY**

- In 1991, Linus Torvalds, a student at the University of Helsinki, Finland, started developing Linux.
- The Linux kernel is written in C language.
- Initially, the name was planned as Freax, but later it became Linux.
- In 1992, Linux was released under the GNU General Public License.

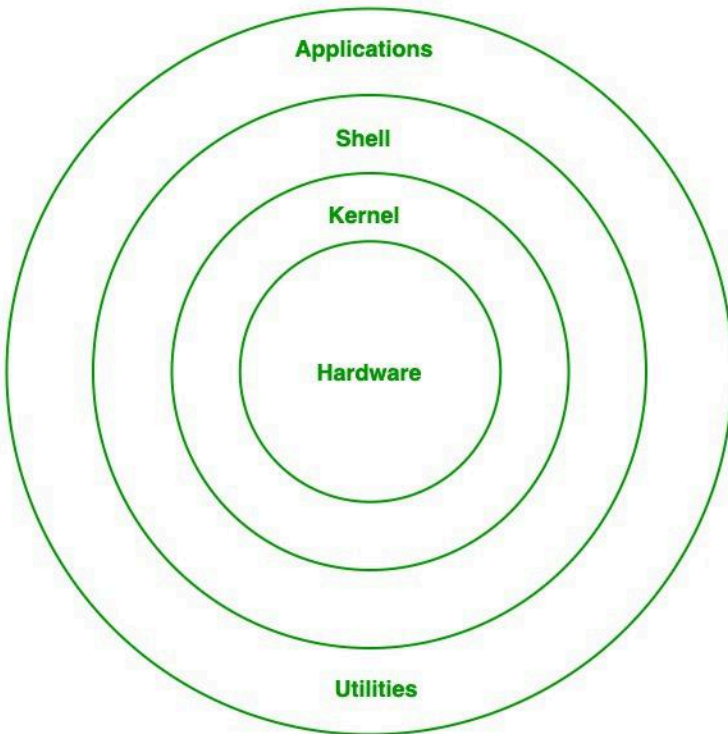
### **OPEN SOURCE**

- Linux is distributed under an open-source license.
- Open source provides:
  - Freedom to use the software
  - Freedom to study and modify the code
  - Freedom to redistribute copies
  - Freedom to distribute modified versions

### **ARCHITECTURE**

The Linux architecture consists of:

- Kernel
- Shell
- Commands
- Terminal



# Introduction to Essential Linux Commands

Linux is a powerful and popular operating system, especially for servers, developers, and advanced users. To effectively use Linux, you interact with it through the **Command Line Interface (CLI)**, also known as the terminal or shell.

This document provides a detailed, yet simple, guide to essential Linux commands.

## Basic Navigation and File Management

These commands help you move around the filesystem and handle files and directories (folders).

| Command            | Simple Language Explanation                                                                                     | Example                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>pwd</code>   | <b>P</b> rint <b>W</b> orking <b>D</b> irectory. Shows you the full path of the directory you are currently in. | <code>pwd</code>                                            |
| <code>ls</code>    | <b>L</b> ist. Shows you the files and folders inside your current directory.                                    | <code>ls</code>                                             |
| <code>ls -l</code> | A "long list" that shows detailed information (permissions, size, owner, date).                                 | <code>ls -l</code>                                          |
| <code>cd</code>    | <b>C</b> hange <b>D</b> irectory. Used to move from one directory to another.                                   | <code>cd Documents</code> (moves into the Documents folder) |
| <code>cd ..</code> | Moves you up one level (to the parent directory).                                                               | <code>cd ..</code>                                          |
| <code>cd ~</code>  | Moves you directly to your user's Home directory (the starting point).                                          | <code>cd ~</code>                                           |
| <code>mkdir</code> | <b>M</b> ake <b>D</b> irectory. Creates a new folder.                                                           | <code>mkdir new_project</code>                              |
| <code>touch</code> | Creates a brand-new, empty file.                                                                                | <code>touch notes.txt</code>                                |

```
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ cd data
bash: cd: data: Not a directory
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ touch app
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ mkdir app
mkdir: cannot create directory 'app': File exists
ubuntu:~$ mkdir app2
ubuntu:~$ cd app2
ubuntu:~/app2$ cd ..
ubuntu:~$ rmdir app
rmdir: failed to remove 'app': Not a directory
ubuntu:~$ ls
a.txt  app  app2  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ nano app
Command 'nano' not found, did you mean:
  command 'mono' from deb mono-runtime (6.8.0.105+dfsg-3.5ubuntu1)
  command 'nano' from deb nano (7.2-2ubuntu0.1)
  command 'nona' from deb hugin-tools (2023.0.0+dfsg-1)
Try: apt install <deb name>
ubuntu:~$ cd app2
ubuntu:~/app2$ nano file
ubuntu:~/app2$ ls
file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ubuntu:~/app2$ vi file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ok new line added here ok
```

## File Operations

Once you have files, you need commands to view, copy, move, and delete them.

| Command            | Simple Language Explanation                                                          | Example                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>cat</code>   | <b>C</b> atenate. Shows the entire content of a file directly in the terminal.       | <code>cat notes.txt</code>                                                                                     |
| <code>less</code>  | Views file content page-by-page, good for large files. Press <code>q</code> to quit. | <code>less very_large_log.txt</code>                                                                           |
| <code>cp</code>    | <b>C</b> opy. Copies a file from one location to another.                            | <code>cp file1.txt /home/&lt;span type="placeholder" placeholder-type="person"&gt;&lt;/span&gt;/backup/</code> |
| <code>mv</code>    | <b>M</b> ove. Moves a file (like cutting and pasting). Also used to rename files.    | <code>mv oldname.txt newname.txt</code>                                                                        |
| <code>rm</code>    | <b>R</b> emove. Permanently deletes a file (be careful, there is no Recycle Bin).    | <code>rm temp.log</code>                                                                                       |
| <code>rm -r</code> | Removes an empty or non-empty directory. The <code>-r</code> stands for recursive.   | <code>rm -r old_folder</code>                                                                                  |

## Getting Help

The Linux command line has built-in help features for almost every command.

| Command          | Simple Language Explanation                                                  | Example             |
|------------------|------------------------------------------------------------------------------|---------------------|
| <code>man</code> | <b>M</b> anual. Shows the comprehensive reference manual page for a command. | <code>man ls</code> |

| Command             | Simple Language Explanation                                                       | Example                |
|---------------------|-----------------------------------------------------------------------------------|------------------------|
| <code>--help</code> | A simple option available on most commands that provides a quick list of options. | <code>ls --help</code> |

## Viewing System Information

These commands provide insights into how your system is running.

| Command               | Simple Language Explanation                                                                               | Example               |
|-----------------------|-----------------------------------------------------------------------------------------------------------|-----------------------|
| <code>date</code>     | Displays the current system date and time.                                                                | <code>date</code>     |
| <code>whoami</code>   | Tells you the name of the user you are currently logged in as.                                            | <code>whoami</code>   |
| <code>uname -a</code> | Displays detailed information about your operating system kernel.                                         | <code>uname -a</code> |
| <code>df -h</code>    | <b>Disk Free.</b> Shows the amount of available disk space, in "human-readable" format.                   | <code>df -h</code>    |
| <code>top</code>      | Shows a real-time list of running processes, like the Windows Task Manager. Press <code>q</code> to quit. | <code>top</code>      |

## Search and Location

Commands to find files or text within files.

| Command           | Simple Language Explanation                                                      | Example                                                                                                     |
|-------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>grep</code> | <b>Global Regular Expression Print.</b> Searches for specific text inside files. | <code>grep "error" server.log</code><br>(finds the word "error" in the log file)                            |
| <code>find</code> | Searches for files and directories based on various criteria (name, size, date). | <code>find . -name "*.txt"</code><br>(finds all files ending in <code>.txt</code> in the current directory) |

## Permissions

Linux uses permissions to control who can read, write, or execute a file.

| Command            | Simple Language Explanation                                      | Example                                                                                                             |
|--------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>chmod</code> | <b>Change Mode.</b> Changes the permissions of a file or folder. | <code>chmod +x script.sh</code> (makes a file executable)                                                           |
| <code>chown</code> | <b>Change Owner.</b> Changes the owner of a file or folder.      | <code>chown &lt;span type="placeholder" placeholder-type="person"&gt;&lt;/span&gt;:users important_file.conf</code> |

For further details on any specific command, please refer to the official documentation or use the `man` command. A list of advanced commands is available in the document titled

📄 File .